

Nguyen Thuy Trang

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Objective

With a background in Mathematics and Computer Science, I aim to pursue research in machine learning, computational methods, and human-centered AI. I am particularly interested in developing data-driven and intelligent systems that address real-world challenges, support human decision-making, and improve human well-being and quality of life. Through hands-on research, I hope to deepen my understanding of these areas and contribute to meaningful interdisciplinary projects in a collaborative academic environment.

Work Experience

Institute of Physics - VAST, Research Assistant Intern May - August 2025

- Processed and cleaned multi-modal sensor signals; extracted features for classification tasks; Researched AI/ML approaches for analyzing sensor data in physics and biomedical applications.
- Assisted in academic writing and preparation of research reports.

Dasens, Developer Feb 2026 – present

- Collaborated remotely with a Canada-based cross-functional team to develop an AI-powered EdTech platform for personalized student assessment and learning analytics.
- Developed backend services and contributed to AI-driven features, including retrieval and workflow automation.
- Technologies: Python, Node.js, React.js, LangChain, LightRAG, n8n.

Education

VNU - University of Science, BS in Mathematics and Computer Science 2022 – now

- GPA: 3.3/4.0
- **Relevant Coursework:** Algebra, Calculus, Probability, Statistics, Data Structures & Algorithms (DSA), Machine Learning, Computer Vision, Natural Language Processing & Apps.

SKILLS

Programming Languages	Python, Java, JavaScript, MATLAB, R, Arduino
AI	ML/DL, NLP (RAG, Knowledge Graph, Agent, LLMs)
Data	SQL, relational database, Spark
Tools & Technologies	Git, GitHub, Docker, Notion, Latex, AI Tools, n8n

LANGUAGES

English	CEFR B2
Japanese	JLPT N4 (currently preparing for N3)

PROJECTS

Gardenia - FlowerVision 2025

- Developed a web-based flower recognition application using Deep Learning, combining ConvNeXt for flower classification and YOLOv8 for automatic flower detection and cropping.
- Implemented real-time image recognition through webcam and image upload, supporting top-3 predictions with confidence scores, large image preview, drag-and-drop upload, and responsive UI design.
- Built a Flask backend with API endpoints for image classification and real-time detection. The inference pipeline follows: input image, YOLO detection, cropped flower region, ConvNeXt classification, and Wikipedia information retrieval to provide detailed flower information.
- Field: Computer Vision, Deep Learning, Image Classification, Object Detection, Web Application Development.

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| AYALEARNING – AI-Powered Japanese Learning Platform | 2025 |
| <ul style="list-style-type: none"> Architected and developed a full-stack AI-powered language learning platform integrating dictionary search, adaptive vocabulary training, interactive practice games (inspired from traditional Japanese games), and real-time conversational AI with animated Live2D characters, enabling users to practice kaiwa with different AI-driven personalities through immersive avatar interaction. Implemented a personal AI assistant agent using LangChain and LangGraph to orchestrate multi-step task execution, including answering questions, generating personalized study plans, scheduling reviews, and guiding user workflows across the platform. Technologies: ReactJS, TypeScript, TailwindCSS, NodeJS, PostgreSQL, LangGraph, LLM APIs, Live2D. | |

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| HomeMind – AI-powered Smart Home Management Application | 2026 |
| <ul style="list-style-type: none"> Developed a smart home management application that enables users to monitor and control IoT devices such as lighting, air conditioners, smart locks, cameras, and environmental sensors through a centralized interface. Features: Real-time device monitoring and control; room-based management; automation routines and scheduling; sensor-based triggers; notifications for abnormal events; energy consumption tracking. AI Agent: Integrated an LLM-based assistant for natural-language device control, automatic routine creation, smart-home status queries, and context-aware recommendations based on sensor data and user preferences. Intelligent automation: Combined environmental data and user behavior to automatically adjust devices, suggest energy-saving actions, and execute multi-device scenarios such as “Good Night”, “Leaving Home”, and “Movie Mode”. Technologies: React/Flutter, FastAPI, MQTT, WebSocket, PostgreSQL, Docker, LLM, LangGraph, Langfuse, ZEP. | |

RESEARCH EXPERIENCES

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| Application of machine learning in knee-injury detection based on IMU sensor data | 2025 |
| <i>The 9th International Conference on Applied & Engineering Physics</i> | |
| <i>Nguyen Thuy Trang</i> , Pham Thi Duyen, Nguyen Trong Hieu, Nguyen Quang Linh, Nguyen Anh Duong, Vu Duong | |
| Building a dataset and information retrieval system from admission counseling forums | 2025 |
| <i>Third Prize, Annual Student Research Conference, MIM, HUS – Apr 2025</i> | |
| Pham Tuan Anh, <i>Nguyen Thuy Trang</i> , Le Thanh Truong, Nguyen Cong Manh | |

CERTIFICATES & AWARDS

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| The 9th Global Competition for Future Design, Southeast Asia Regional Finals | July 2026 |
| <i>Best Technologically Impressive Award</i> | |
| Global Program on Innovation and Entrepreneurship | Sep - Nov 2025 |
| <i>Tsinghua University, China</i> | |
| Guangxi University International Students AI Winter Camp 2025 | Dec 2025 |
| <i>Guangxi University, China</i> | |
| VIASM Summer School “Research Experience for Undergraduates” | 2025 |
| <i>Vietnam Institute for Advanced Studies in Mathematics, Vietnam</i> | |
| Summer School of Agentic AI | 2025 |
| <i>Naver & Hanoi University of Science and Technology (HUST)</i> | |
| Samsung Innovation Campus - AI Track | 2024 |

EXTRACURRICULAR ACTIVITIES

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| HUS Applied Mathematics and Informatics Club, Head of Events | 2023 – 2025 |
| ULIS Japanese Club, member | 2023 – 2025 |
| Math Open Day, volunteer | 2024 |